## Compiled Solutions – Senior Division

### 1. 10-11 C1 What Does this Program Do?

A * B + C * D – E = 9 * 6 + 1 * 8 – 18 = 44 44

| A | B | C | D | E |
|---|---|---|---|---|
| 2 | 4 | 1 | 3 | 5 |
| 2 | 5 | 1 | 3 | 5 |
| 2 | 5 | 1 | 8 | 5 |
| 13 | 5 | 1 | 8 | 5 |
| 13 | 5 | 1 | 8 | 18 |
| 8 | 5 | 1 | 8 | 18 |
| 8 | 6 | 1 | 8 | 18 |
| 9 | 6 | 1 | 8 | 18 |

### 2. 10-11 C2 What Does this Program Do?

S counts the numbers from 1 to the square root of I that divide into I. 19

| I | 1 | 2 | 3 | 4 | 4 | 5 | 6 | 6 | 7 | 8 | 8 | 9 | 9 | 10 | 10 | 11 | 12 | 12 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| J | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 3 | 1 | 2 | 1 | 1 | 2 | 3 |
| S | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |

### 3. 11-12 C1 What Does this Program Do?

The table contains the values of A, B, C, and D after each line. 3

| A | B | C | D |
|---|---|---|---|
| 2 | 10 | 5 | 20 |
| 4 | 10 | 5 | 20 |
| 4 | 10 | 6 | 20 |
| 4 | 10 | 4 | 20 |
| 4 | 10 | 4 | 6 |
| 4 | 10 | 4 | 7 |
| 4 | 6 | 4 | 7 |
| 4 | 6 | 4 | 7 |
| 4 | 6 | 4 | 3 |
| 4 | 6 | 4 | 4 |

### 4. 11-12 C3 What Does this Program Do?

This program alters the entries by taking differences. Then it checks for divisibility by 2 and then by 4. Finally, it checks to see if the entry is a perfect square. If a test fails then 1 is added or it is squared. There are only 4 which are not 0. 4

| | | | |
|---|---|---|---|
| 0 | 9 | 0 | 1 |
| 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 |
| 0 | 361 | 0 | 25 |

**5. 12-13 C1 What Does this Program Do?**

The table contains the values of a, b, c, d and e after each line. 7

| a | b | c | d | e |
|---|---|---|---|---|
| 18 | 8 | 2 | -1 | 0 |
| 18 | 8 | 2 | -1 | 8 |
| 18 | 8 | 2 | -2 | 8 |
| 18 | 22 | 2 | -2 | 8 |
| 18 | 22 | 2 | -2 | 4 |
| 18 | 22 | 2 | -2 | 4 |
| 18 | 22 | 1 | -2 | 4 |

So (b + d) / e + ( a + e) / b + c = (22 -2)/4+(18+4)/22+1 = 7

**6. 12-13 C3 What Does this Program Do?**

This program counts the number of primes from 2 to 25. 9
They are 2, 3, 5, 7, 11, 13, 17, 19, and 23. There are 9 of them.
It uses the method called the Sieve of Eratosthenes.

**7. 13-14 C1 What Does this Program Do?**

The table contains the values of a, b, c, d and e after each line. -18

| a | b | c | d | e |
|---|---|---|---|---|
| 10 | 5 | 20 | 1 | 2 |
| 10 | 5 | 20 | 1 | 2 |
| 10 | 5 | 20 | 25 | 2 |
| 10 | 20 | 20 | 25 | 2 |
| 10 | 8 | 20 | 25 | 2 |
| 10 | 8 | 20 | 5 | 2 |

So c / (b + e) – d ^ 2 + a / e = 20 / (8 + 2) – 5^2 + 10/2 = -18

**8. 13-14 C3 What Does this Program Do?**

This program changes all negative entries to 0, 3
then replaces all the even entries by the result of dividing by 2,
and finally makes all the entries less than 3 equal to 0.
That leaves 3 that are not 0.

Original:
-1 0 1 2
-2 0 2 4
-3 0 3 6
-4 0 4 8

Final:
0 0 0 0
0 0 0 0
0 0 3 3
0 0 0 4

**9. 14-15 C1 What Does this Program Do?**

The table contains the values of a, b, c, d and e after each line. -26

| a | b | c | d | e |
|---|---|---|---|---|
| 48 | 12 | 0 | -2 | 3 |
| 4 | 12 | 0 | -2 | 3 |
| 4 | 3 | 0 | -2 | 3 |
| 4 | 3 | 0 | 4 | 3 |
| 1 | 3 | 0 | 4 | 3 |
| 1 | 3 | 0 | 2 | 3 |

So a * (b + c) / e – b ^ e – c / (d + 1)
= 1 * (3 + 0) / 3 – 3 ^ 3 – 0 / (2 + 1) = 1 – 27 = -26

**10. 14-15 C3 What Does this Program Do?**

This program changes all the values in the original table and prints the entry with the largest value. The final values are: 24

| | | | |
|---|---|---|---|
| 10 | 15 | 24 | 15 |
| 13 | 11 | 12 | 13 |
| 4 | 2 | 7 | 10 |
| 5 | 2 | 7 | 6 |

**11. 15-16 C1 What Does this Program Do?**

The table contains the values of a, b, c, d and e after each line. -17

| a | b | c | d | e |
|---|---|---|---|---|
| 16 | 2 | 4 | 0 | 5 |
| 14 | 2 | 4 | 0 | 5 |
| 14 | 2 | 4 | 0 | 5 |
| 14 | 2 | 4 | 1 | 5 |
| 10 | 2 | 4 | 1 | 5 |
| 5 | 2 | 4 | 1 | 5 |
| 5 | 2 | 4 | 0 | 5 |

So a ^ d + b * e / a – c * (a / e + b ^ b) =
5 ^ 0 + 2 * 5 / 5 – 4 * (5 / 5 + 2^ 2) = 1 + 2 – 4*5 = -17

**12. 15-16 C3 What Does this Program Do?**

This program changes all the values in the original table and counts how many are 0. The final values are: 11

| | | | |
|---|---|---|---|
| 0 | 9 | 0 | -9 |
| -6 | 0 | -5 | 0 |
| 0 | 0 | 0 | 0 |
| 7 | 0 | 0 | 0 |

4. **02-03 C4 What Does This Program Do – Strings**

| I | B |
|---|---|
| 0 | C |
| 2 | CNNC |
| 4 | CNNCE |
| 6 | ECNNCEE |
| 8 | ECNNCEELLECNNCEE |

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8. **03-04 C4 What Does this Program Do – Strings**

Y = M

The original A has its letters compared from each end, letter 1 is compared with letter 21, 2 is compared with 20, etc. If the first numbered letter is the bigger, it is added to X. The second loop adds just those letters less than N to X= "UNTSTTOM" and Y = "M".

12. **04-05 C4 What Does this Program Do – Strings**

| **I** | **A** |
|---|---|
| 2 | E |
| 3 | AE |
| 5 | EEAE |
| 6 | EEAEE |
| 11 | EEEAEE |
| 15 | EEEEAEE |
| 16 | AEEEEAEE |
| 18 | AAEEEEAEE |
| 23 | AAAEEEEAEE |
| 24 | AAAEEEEAEEL |

AAAEEEEAEEL